

MULTI-TEACHER ON-POLICY DISTILLATION: TASK WEIGHTING AND SPARSE PARAMETER UPDATES

Anonymous authors

Paper under double-blind review

ABSTRACT

Multi-teacher on-policy distillation (MOPD) combines specialist feedback in one student. We study how loss reduction and sparse parameter updates relate to capability integration through three empirical questions. First, we analyze how token averaging determines cross-domain weights and within-domain response weighting. An exact finite-batch decomposition provides an explanation to test against capability comparisons. Second, building on established OPD sparsity, we examine sparse changes in single-teacher and multi-teacher training, measure overlap between teacher-conditioned updates at the same MOPD checkpoint, and ask whether larger teacher distribution differences correspond to larger differences between these update supports. Teacher–student distance is recorded alongside the teacher-pair comparison. Third, we compare sampled-token policy-gradient and top- k losses in both single-teacher and multi-teacher OPD. This final comparison tests whether update sparsity materially changes under the two practical losses; further mechanism analysis is conditional on observing a stable difference. **[TODO: Research draft: the proposed experiments remain to be completed. Replace this sentence with measured findings and their scope.]**

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) transfers domain specialists into one student by scoring student-generated responses with their assigned teachers (Ma et al., 2026). Its intended outcome is balanced capability integration. Open-MOPD identifies capability imbalance in a controlled routed setting and studies token-share weighting, changing distillation gaps, and stale student-dependent rewards (Gao et al., 2026). These findings motivate examining both the loss weights given to each domain and the parameter updates induced by its teacher.

We organize the paper around three studies. The first examines loss reduction as an explanation for imbalance. A global token mean gives long-response domains greater weight. Averaging tokens separately within each domain changes those domain weights, but still weights longer responses more heavily inside the domain. We derive these effects and test their relationship to multi-task capability. This analysis does not require a new sampling or weighting algorithm.

The second study is the empirical center of the paper. Prior work already reports sparse parameter changes in RL (Mukherjee et al., 2025) and in OPD (Yu et al., 2026; Shen et al., 2026), including support reuse when varying the teacher. We first verify update sparsity in our single-teacher and multi-teacher settings. We then ask whether different teachers acting on the *same MOPD student* update overlapping parameter subsets. Finally, we test whether teacher pairs with larger distribution differences induce more different update supports. Teacher–student distance is measured as a related factor that may help explain transfer and update differences. Reproducing existing sparsity results is a foundation for this study, not a claim of first discovery.

The third study is a focused comparison of loss choices. We compare sampled-token policy-gradient and top- k losses in both single-teacher and multi-teacher OPD, keeping the other settings comparable. If they show no material sparsity difference in the tested regime, that is a reportable observation and the study stops at that comparison. If a stable difference appears, additional controls can examine its source. We do not make estimator-noise analysis the main contribution or assume that sampled-token supervision must produce sparser parameter updates.

The intended contributions are therefore:

1. An empirical and algebraic explanation of how loss reduction and length weighting affect capability balance across tasks.
2. A study of sparse OPD updates, the overlap of teacher-conditioned supports within MOPD, and their relationship to teacher–teacher and teacher–student distribution differences.
3. A comparison of parameter-update sparsity under sampled-token PG and top- k losses in single-teacher and multi-teacher distillation.

The core questions permit positive, negative, and null relationships. Support overlap does not by itself imply harmful interference, and a sparsity difference does not by itself justify choosing a loss. Capability and signed cross-task effects provide context for these measurements. **[TODO: Replace the prospective empirical contributions with observed findings and their measured scope once the experiments are complete.]**

2 SETTING AND MEASUREMENT LEVELS

There are D domains with prompt distributions $\mathcal{D}_d$, frozen teachers q_d , and a shared student p_θ . A student-generated response y contains T valid positions with prefixes $h_t = (x, y_{<t})$. Unless testing a different objective, the fixed-prefix reference is the reverse-KL loss

$$\ell_d(\theta; h) = \sum_{v \in \mathcal{V}} p_\theta(v | h) \log \frac{p_\theta(v | h)}{q_d(v | h)}. \quad (1)$$

Student and teachers share a tokenizer and vocabulary. Prefixes, teachers, and loss weights are held fixed when differentiating a local update. This derivative is not asserted to be the full gradient of a sequence-level objective whose state distribution depends on θ .

We distinguish the raw gradient g_d , the realized optimizer step u_d , the cumulative displacement $\Delta_d = \theta_d - \theta_0$, and downstream capability $C_d(\theta)$. Sparsity of one object does not establish sparsity of the others. A vocabulary alternative, a response position, and a parameter coordinate are also different units.

At a common checkpoint, independently applying each teacher to copies of the same student makes its local effects comparable. Separately trained OPD students provide observed single-domain references for online experiments. They are neither upper bounds nor guarantees of joint attainability. Capability, loss on a common bank, and loss on fresh student prefixes are reported separately. A decrease in average KL alone does not establish balanced capability integration.

3 LENGTH NORMALIZATION CHANGES THE TRAINING MIXTURE

Length imbalance depends on how token losses are reduced. MOPD writes a response-normalized objective (Ma et al., 2026), whereas Open-MOPD analyzes a token-mean implementation in which instruction following supplies approximately 20% of prompts but only 1% of response tokens (Gao et al., 2026). This observation motivates separating the training objective from the sampling mixture. It does not establish a length pathology intrinsic to multi-teacher distillation. The formal objective and each implementation’s complete reduction path, including gradient accumulation, require separate verification. **[TODO: Reduction audit.]**

Three reduction orders. Fix a rollout batch containing $n_d \geq 1$ responses from domain d . Response i has $T_{di} > 0$ valid positions, position losses ℓ_{dit} , and mean loss $\bar{\ell}_{di} = T_{di}^{-1} \sum_t \ell_{dit}$. Write $N_d = \sum_i T_{di}$, $N = \sum_d N_d$, and $s_d = N_d/N$. The target domain weights $a_d \geq 0$, with $\sum_d a_d = 1$,

are distinct from the prompt proportions $p_d = n_d / \sum_j n_j$. Consider

$$L_{\text{tok}} = \frac{1}{N} \sum_{d,i,t} \ell_{dit} = \sum_d s_d \frac{\sum_i T_{di} \bar{\ell}_{di}}{N_d}, \quad (2)$$

$$L_{\text{DT}} = \sum_d a_d \frac{\sum_i T_{di} \bar{\ell}_{di}}{N_d}, \quad (3)$$

$$L_{\text{DR}} = \sum_d a_d \frac{1}{n_d} \sum_i \bar{\ell}_{di}. \quad (4)$$

These respectively average tokens globally, tokens within domains, and positions within responses before averaging responses within domains. Multiplying each domain’s token loss by a_d/s_d converts Equation 2 into Equation 3; this is Open-MOPD’s token-share balancing. It preserves length weighting within each domain. An ordinary mean over response means instead equals Equation 4 with $a_d = p_d$.

Proposition 1 (Fixed-batch length–gradient decomposition). *Hold the rollouts, masks, and domain weights fixed when differentiating; any surrogate advantages are detached. Define $h_{di} = T_{di}^{-1} \sum_t \nabla_{\theta} \ell_{dit}$, $\bar{h}_d = n_d^{-1} \sum_i h_{di}$, and $\bar{T}_d = n_d^{-1} \sum_i T_{di}$. Let*

$$C_d = \frac{1}{n_d} \sum_i (T_{di} - \bar{T}_d)(h_{di} - \bar{h}_d). \quad (5)$$

Then

$$\nabla L_{\text{tok}} = \sum_d s_d \left(\bar{h}_d + \frac{C_d}{\bar{T}_d} \right), \quad (6)$$

$$\nabla L_{\text{DT}} = \sum_d a_d \left(\bar{h}_d + \frac{C_d}{\bar{T}_d} \right), \quad \nabla L_{\text{DR}} = \sum_d a_d \bar{h}_d. \quad (7)$$

Proof. Expanding the empirical covariance gives $n_d^{-1} \sum_i T_{di} h_{di} = \bar{T}_d \bar{h}_d + C_d$. Divide by $\bar{T}_d$ and substitute into each reduction. $\square$

Domain token normalization therefore removes the coefficient mismatch $s_d - a_d$ while retaining the within-domain length–gradient covariance. These are standard weighted-mean identities, used here to explain existing training recipes rather than introduce a new algorithm. Equal token counts do not imply equal gradient norms or compatible update directions; neither token shares nor scalar reward magnitudes capture vector cancellation or optimizer preconditioning. The identities are exact for a fixed batch. In expectation, a sample ratio generally differs from $\mathbb{E}[Th]/\mathbb{E}[T]$; the latter describes an appropriate large-batch limit, not an exact finite-batch expectation.

Controlled observations. To test this explanation, compare all three reductions on identical rollouts, teacher scores, and surrogate signals at early, middle, and late checkpoints. Measure s_d , $\bar{h}_d$, $C_d/\bar{T}_d$, and gradient directions before clipping. First set $a_d = p_d$, or use equal prompt quotas, to isolate length weighting; evaluate uniform target weights separately. Otherwise, correcting length also changes the domain prior. **[TODO: Gradient decomposition.]** Follow these measurements with matched training branches that retain the same prompt schedule, response limits, teachers, and update settings, while generating their own on-policy responses. Report capability curves against both updates and cumulative generated tokens. **[TODO: Training curves.]** Reducing the response limit is a separate intervention: it changes the available prefixes and may remove solution endings. Its effect cannot be identified with loss reweighting, nor do the identities imply that response normalization always improves capability integration.

4 SPARSE OPD UPDATES AND TEACHER SUBNETWORK OVERLAP

We investigate three questions in sequence: whether OPD updates are sparse, whether different teachers update overlapping subnetworks of the same MOPD student, and whether larger teacher distribution differences correspond to larger subnetwork differences. The following experiments are planned; their outcomes are not yet established.

4.1 DO SPARSE OPD UPDATES PERSIST UNDER MULTIPLE TEACHERS?

Sparse parameter changes have been reported for RL fine-tuning (Mukherjee et al., 2025) and OPD (Yu et al., 2026; Shen et al., 2026). We first reproduce this measurement in our setting and extend the comparison to MOPD. From the same initialization θ_0 , we measure single-teacher deltas $\Delta_{i,t}^{\text{single}}$ and the joint delta $\Delta_t^{\text{joint}} = \theta_t - \theta_0$, comparing training curves separately against domain exposure and compute. For an n -coordinate delta v ,

$$s_\tau(v) = \frac{1}{n} \sum_{a=1}^n \mathbf{1}\{|v_a| \leq \tau\}. \quad (8)$$

We report threshold sweeps, update norms, and top-coordinate energy concentration using FP32 master-weight deltas, with bf16 checkpoint measurements for comparison. This distinguishes small accumulated changes from rounding-induced zeros; endpoint sparsity does not establish that instantaneous gradients are exactly sparse. This first experiment is a replication and extension, not a claim to discover OPD sparsity.

4.2 WHICH SUBNETWORKS DO TEACHERS UPDATE IN THE SAME STUDENT?

At selected checkpoints θ_t of one MOPD run, each teacher processes its routed student-rollout batch independently. Every diagnostic branch starts from the identical student and optimizer state. We record each teacher’s gradient $g_{i,t}$ and virtual optimizer displacement $u_{i,t}$ without advancing the shared training run. These local probes describe each teacher’s update tendency at the actual joint-training checkpoint; they do not decompose AdamW’s joint update additively.

For either probe $v_{i,t}$, define its threshold support and pairwise Jaccard overlap as

$$A_{i,t}^\tau = \{a : |v_{i,t,a}| > \tau\}, \quad J_{ij,t} = \frac{|A_{i,t} \cap A_{j,t}|}{|A_{i,t} \cup A_{j,t}|}. \quad (9)$$

We also compare masks retaining the same top fraction of coordinates, globally and within layers, against density-matched random masks. Threshold masks reflect update scale as well as location; fixed-density masks emphasize location. Raw-gradient masks check whether shared momentum or decay dominates virtual-step overlap. Independent single-teacher endpoint supports provide a secondary comparison, but cannot identify teacher contributions along the MOPD trajectory. Signed alignment is an auxiliary diagnostic: overlapping coordinates can support compatible or opposing updates, so low overlap is not assumed preferable.

4.3 DO MORE DIFFERENT TEACHERS UPDATE MORE DIFFERENT SUBNETWORKS?

At each probe checkpoint, all teachers score the same cached student-prefix bank $\mathcal{C}_t$. With q_i denoting teacher i ’s distribution, measure

$$\begin{aligned} D_{ij,t}^{\text{TT}} &= \mathbb{E}_{c \sim \mathcal{C}_t} \text{JS}(q_i(\cdot | c), q_j(\cdot | c)), \\ D_{i,t}^{\text{TS}} &= \mathbb{E}_{c \sim \mathcal{C}_t} \text{JS}(q_i(\cdot | c), p_{\theta_t}(\cdot | c)). \end{aligned} \quad (10)$$

We plot teacher-pair divergence against subnetwork distance $1 - J_{ij,t}$ at the same checkpoint, recording both teacher–student gaps as auxiliary descriptors. A small common-context check recomputes all teachers’ masks on identical prefixes, assessing whether routed-domain differences explain the observed association. Code and science currently share one Qwen3-4B teacher; comparisons between these routes measure context differences, not differences between independent teachers. We report pair trajectories and run-level uncertainty without treating shared teachers or repeated checkpoints as independent observations. A positive distance–difference relationship is a hypothesis to test, not a consequence of sparsity or an assumption about optimal MOPD behavior.

5 DOES LOSS DESIGN CHANGE UPDATE SPARSITY?

Our third study is a focused comparison of sampled-token policy gradients (PG) and top- K distillation in single-teacher and multi-teacher settings (Table 1). Prior work already observes concentrated

Setting	Sampled-token PG	Top- K distillation
Single teacher	Separate OPD run	Matched separate OPD run
Multiple teachers	Joint MOPD run	Matched joint MOPD run

Table 1: Primary loss-design comparison. Both columns use the same sparsity measurements and capability evaluations.

OPD parameter changes under dense feedback (Yu et al., 2026). We ask whether these two practical loss choices produce materially different update sparsity under otherwise matched configurations, without assuming that sampled supervision is sparser or reduces interference.

Within each setting, paired runs share the student initialization, teachers, prompt schedule, length limits, task normalization, optimizer configuration, and training steps. We fix and document K , support selection, and probability normalization for the chosen top- K implementation (Ma et al., 2026). Online rollouts follow each run’s evolving student; generated-token exposure and compute are reported alongside matched-step comparisons.

We measure actual optimizer-step sparsity and cumulative parameter changes at identical check-points, using the same FP32 measurement procedure and thresholds across conditions. Update norms and scale-normalized concentration provide context for threshold-visible sparsity. Repeated seeds yield effect estimates and confidence intervals. Held-out task scores establish whether a sparser update accompanies comparable learning; sparsity alone does not demonstrate better capability integration.

At a fixed prefix, fresh sampled-token PG with detached log-ratio coefficients has the same expected gradient as full-vocabulary reverse KL, a known identity (Yu et al., 2026; Oh et al., 2026). This does not equate practical top- K objectives, length-weighted original rollout samples, nonlinear Adam updates, or long-term parameter supports.

If confidence intervals fall within a prespecified materiality margin, we will report no material sparsity difference within the tested regime and stop this line of analysis; intervals too wide to resolve that margin are inconclusive. Only a stable, meaningful gap would motivate an optional follow-up separating sampling noise from truncation bias, for example using a matched full-vocabulary reference or a variance-reduced sampled estimator. These mechanism controls are conditional extensions, not required components of the primary study, and no preference for sampled PG is assumed.

6 EXPERIMENTAL DESIGN

6.1 SHARED SETUP AND SCOPE

The working setup retains a Qwen3-1.7B student and four domains: mathematics, code, instruction following, and science. Math and IF use domain-specific Qwen3-1.7B RL teachers. Code and science currently share one Qwen3-4B teacher as a temporary substitute for separate specialists. There are thus four routed domains but only three distinct teacher weight sets. The repeated teacher is a useful context comparison, not two independent observations of teacher diversity.

The online anchor uses equal domain prompt quotas and weights, 16 micro-batches of four responses per update, one optimizer update per fresh rollout batch, and a 4,096-token reference cap. The working budget is at most 500 updates per run, subject to a measured throughput pilot on the available 96-GB learner and 48-GB rollout/teacher GPUs. Exact revisions, optimizer and decoding settings, data splits, and evaluation versions remain to be frozen. The repository does not yet contain measured outcomes for this study.

The core design reuses runs across questions. A representative single-teacher setting is selected before examining results; it and the joint setting each compare PG with top-64. The joint top-64 run with domain-response averaging also anchors the normalization comparison, which adds global-token and domain-token averaging. This gives six training configurations for an initial study, not a requirement to train a new expert for every teacher pair. Additional single-teacher settings and repeated seeds broaden the scope only as resources allow.

6.2 STUDY 1: LENGTH WEIGHTING AS AN EXPLANATORY OBSERVATION

Use the same top-64 loss, teachers, prompt quotas, and initialization for all three reductions in Section 3. First compare them on identical cached batches: record response counts, lengths, domain token shares, response-mean gradients, and the covariance correction. Verify the finite-batch decomposition and measure each proposed update on a common held-out bank. A mean of unequal micro-batch means need not equal the intended batch mean, so the implementation records denominators across accumulation and distributed workers.

Then compare the three online training branches and their per-domain capability. Their prompt schedules are paired, but each generates its own on-policy responses. Equal prompt quotas separate length weighting from changes in the desired domain prior. An unequal-quota diagnostic first sets target weights to prompt shares before separately testing uniform weights. A 1,024 versus 4,096 cap comparison is a supporting length observation, not the paper’s new algorithm. Report truncation and answer-completion rates and distinguish exposure from compute.

6.3 STUDY 2A: VERIFY SPARSITY IN SINGLE-TEACHER AND MULTI-TEACHER OPD

For the representative single-teacher and joint runs, save the initial, early, intermediate, and final student checkpoints. Measure cumulative parameter displacement from each run’s own common initialization and report threshold curves and update-energy concentration. Repeat using FP32 master weights when available alongside exported-checkpoint measurements. The comparison establishes the observed sparsity regime for our experiments and checks whether it persists under joint training; it does not claim to discover OPD sparsity for the first time.

6.4 STUDY 2B: MEASURE TEACHER-CONDITIONED OVERLAP INSIDE MOPD

At each selected *joint* checkpoint θ_t , freeze the student and optimizer state. For each routed domain, draw a diagnostic batch from that checkpoint’s student policy and compute its teacher-conditioned raw gradient and proposed optimizer step on a separate copy. Compare their supports at matched coordinate density, retaining both same-layer random baselines and per-layer results. Repeat over diagnostic batches so a single noisy estimate is not mistaken for a stable subnetwork.

The primary figure is a teacher/domain overlap matrix at several MOPD checkpoints, accompanied by each update’s concentration. Independent single-teacher endpoint supports can be shown as a reference, but are not substituted for contributions within the joint student: the online students follow different trajectories. A joint checkpoint delta is also not assumed to decompose uniquely into teacher-specific deltas. Gradient and proposed-step masks are labeled accordingly.

6.5 STUDY 2C: RELATE TEACHER DISTANCE TO SUBNETWORK DIFFERENCE

Score every available teacher on a common held-out student-prefix bank. For each pair, compare its distribution distance with the corresponding support difference, such as $1 - J_{ij}$, at the same checkpoint. Record both teachers’ distances to the current student and their update sizes. Teacher distribution distance must be evaluated on common contexts; comparing distributions conditioned on different prompts does not isolate a teacher difference.

The main analysis relates these common-context teacher distances to *routed* teacher-conditioned supports from Study 2b. A smaller same-prefix update comparison shows whether a relation persists when context is also matched for the updates. This supporting check helps interpret teacher and domain effects without expanding the paper into a full causal study. Available intermediate teacher checkpoints can supply additional distance variation without new teacher training. If the teacher pool remains small, show the individual pairs and report a setting-specific relationship rather than a universal monotone law. Pairs sharing a teacher or training lineage are not independent samples.

The intended figure plots teacher-pair distance against support difference, with checkpoints, domains, and student-relative gaps identified. Larger, smaller, or unchanged support differences are all valid outcomes. Teacher–student gap is additionally compared with update concentration and observed transfer, but no fixed functional relationship is assumed in advance.

6.6 STUDY 3: A FOCUSED PG VERSUS TOP-K COMPARISON

Use the representative single-teacher and joint settings in the same two-by-two comparison: sampled rollout-token PG versus corrected teacher top-64. All use domain-response averaging, matching initialization, optimizer, masks, cap, prompt schedule, and update budget. Document full-vocabulary probability normalization and any clipping. The practical loss comparison is the primary experiment; different objectives are not mislabeled as an isolated causal change in density.

At the same checkpoints, compare cumulative-delta sparsity, update energy concentration, and, in the joint setting, teacher-conditioned support overlap. Report actual step size and capability alongside sparsity, and plot both exposure and compute axes. If the effects are small and consistent with no material difference at the measurement resolution, report that observation with its uncertainty and scope. A stable, meaningful difference can motivate the optional controls in Appendix A.2; they are not prerequisites for the core paper and are not run merely to explain an absent effect.

6.7 EVALUATION AND UNCERTAINTY

Use MATH-500 greedy pass@1, a frozen LiveCodeBench pass@1 slice, IFBench strict accuracy, and GPQA-Diamond average@4. Record evaluator, model, and data revisions and use fixed decoding across compared checkpoints. Report raw per-domain scores and the worst-domain change with the macro-average. Independently trained OPD references, where available, are compared at matched domain exposure; lack of transfer in a single-teacher setting is not itself a multi-teacher failure.

Measure generated tokens, domain responses, updates, and GPU hours separately. Teacher scoring and diagnostics have recorded overhead. Key online comparisons should have repeated paired training seeds; initial one-seed runs are explicitly exploratory. Prompt bootstrap measures evaluation uncertainty conditional on the models, not training-seed variation. A failure to detect a difference is not by itself proof of exact equality; report effect sizes and the range the measurements can distinguish. No numeric sparsity advantage or capability improvement is claimed before measurement.

[TODO: Add the normalization/capability figure, single-versus-joint sparsity curves, teacher-conditioned overlap matrices, teacher-distance scatter plots, and PG/top- k comparison table after collecting data.]

7 RELATED WORK

Capability integration and task weighting. MOPD consolidates routed domain teachers in policy space (Ma et al., 2026). Open-MOPD diagnoses token-share and reward-related imbalances (Gao et al., 2026). Our first study audits the reduction used by each implementation and separates cross-domain weights from within-domain length weighting. The algebra explains existing objectives; it is not introduced as a new allocation algorithm. Prior work on multi-task loss weighting and gradient conflict further motivates distinguishing weight balance from joint capability gains (Chen et al., 2018; Yu et al., 2020).

Sparse updates and teacher compatibility. Mukherjee et al. (2025) find sparse cumulative parameter changes in RL and substantial support reuse across settings. Yu et al. (2026) directly establish related OPD signatures, including teacher-varied overlap and functional masked-training tests. Shen et al. (2026) examine update subspaces and report rank robustness to response-position thinning. These observations are starting points for our study, not discoveries claimed here. Teacher compatibility also affects distillation success (Li et al., 2026; Ma et al., 2026). We first verify sparse updates in our setting, then study overlap between teachers acting on the same MOPD student and its relationship to their distribution differences.

Loss support and gradient estimation. MOPD provides sampled-token and corrected teacher-top- k objectives (Ma et al., 2026). Variance-reduced sampled OPD (Oh et al., 2026), exact-head plus sampled-tail estimation (Zhang & Ba, 2026), and tail-aware losses (Huang & Wei, 2026) show why token count alone does not identify the estimation or optimization problem. Our primary loss study compares parameter sparsity under practical PG and top- k objectives in single-teacher and

multi-teacher OPD. These established controls are reserved for interpreting a stable difference; we do not propose a new estimator.

8 DISCUSSION

The paper first explains loss weighting, then examines sparse updates inside MOPD, and finally compares two practical losses. Its empirical center is the progression from observing sparsity to measuring teacher-conditioned support overlap and relating support differences to teacher distribution differences. Teacher–student distance and signed cross-task effects aid interpretation without replacing those questions.

The PG/top- k comparison is deliberately bounded. Similar sparsity in both single-teacher and multi-teacher settings is a useful observation within the tested regime; it does not require an additional mechanism program. A stable difference can motivate further analysis, but a smaller support alone does not establish better capability integration. Likewise, teacher divergence may or may not predict support difference, and overlap may contain either shared or opposing corrections.

[TODO: State the supported observations, including null relationships, and their model, task, teacher, loss, and measurement scope. Do not convert existing sparsity findings or local algebra into new universal claims.]

AI USE STATEMENT

Generative AI tools assisted with manuscript organization, language, mathematical analysis, and experimental design. The authors remain responsible for reviewing claims, proofs, code, and results.

ETHICS STATEMENT

The proposed study uses existing language models and reasoning and instruction-following datasets. Model and data biases and automated verifier errors can affect capability comparisons. Data provenance, invalid outputs, and evaluator failures should accompany the results.

REPRODUCIBILITY STATEMENT

Sections 3–6 specify the three studies, their local controls, and the distinction between measured updates and downstream capability. Appendix A.3 gives artificial-target controls; Appendix A.1 specifies required records. The repository currently contains the manuscript, analytical calculations, and study specifications. It does not yet contain measured results for the restructured experiments. **[TODO: Release runnable training and diagnostics, frozen configuration, source data for figures, per-seed outputs, exact checkpoints, and measured resource use before claiming empirical reproducibility.]**

REFERENCES

- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 794–803, 2018. URL <https://proceedings.mlr.press/v80/chen18a.html>.
- Huan-ang Gao, Haohan Chi, Yong Yan, Shiyuan Feng, Hanlin Wu, Zheng Jiang, Bingxiang He, Wei-Ying Ma, Ya-Qin Zhang, and Hao Zhou. Open-MOPD: Diagnosing and fixing capability imbalance in multi-teacher on-policy distillation, 2026. URL <https://arxiv.org/abs/2608.19098>.
- Huipeng Huang and Hongxin Wei. Tail-aware top- k on-policy distillation, 2026. URL <https://arxiv.org/abs/2608.14728>.
- Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan-ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large

language models: Phenomenology, mechanism, and recipe, 2026. URL <https://arxiv.org/abs/2604.13016>.

Wenhan Ma, Jianyu Wei, Liang Zhao, Hailin Zhang, Bangjun Xiao, Lei Li, Qibin Yang, Bofei Gao, Yudong Wang, Rang Li, Jinhao Dong, Zhifang Sui, and Fuli Luo. MOPD: Multi-teacher on-policy distillation for capability integration in llm post-training, 2026. URL <https://arxiv.org/abs/2606.30406>.

Sagnik Mukherjee, Lifan Yuan, Dilek Hakkani-Tur, and Hao Peng. Reinforcement learning finetunes small subnetworks in large language models, 2025. URL <https://arxiv.org/abs/2505.11711v2>. Version 2; NeurIPS 2025.

Minjae Oh, Sangjun Song, Gyubin Choi, Yunho Choi, and Yohan Jo. KL for a KL: On-policy distillation with control variate baseline, 2026. URL <https://arxiv.org/abs/2605.07865>.

Zhennan Shen, Yanshu Li, Qingyu Yin, Chak Tou Leong, Zhilin Wang, Yanxu Chen, Rongduo Han, Sunbowen Lee, and Yi R. Fung. On the geometry of on-policy distillation, 2026. URL <https://arxiv.org/abs/2606.07082v3>. Version 3, June 14, 2026.

Guo Yu, Hao-Xuan Ma, Jun-Peng Jiang, Han-Jia Ye, Wenlin Liu, and Yulan Hu. Dense supervision, sparse updates: On the sparsity and geometry of on-policy distillation, 2026. URL <https://arxiv.org/abs/2606.13657v3>. Version 3, July 30, 2026.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning, 2020. URL <https://arxiv.org/abs/2001.06782>.

Lunjun Zhang and Jimmy Ba. EMA policy gradient: Taming reinforcement learning for LLMs with EMA anchor and top-k KL, 2026. URL <https://arxiv.org/abs/2602.04417>.

A MEASUREMENT DETAILS AND OPTIONAL FOLLOW-UP

A.1 TEACHER-CONDITIONED MASKS AT A COMMON JOINT CHECKPOINT

The diagnostic record contains the joint student checkpoint, optimizer state, teacher identity, routed domain, prompt IDs, sampled response and mask, loss reduction, and gradient accumulation denominator. Each teacher-conditioned gradient and proposed step is computed from the same restored student state. They are diagnostic update supports, not a unique historical attribution of the cumulative joint delta.

Record raw and clipped gradients separately. Retain the actual optimizer step and its momentum and decay settings. A zero-gradient optimizer branch can identify motion caused by stored momentum or weight decay when those components dominate a support comparison. Prefer FP32 master-weight differences for cumulative updates when available; converting already rounded BF16 checkpoints to FP32 does not recover unrecorded updates. Record exclusions, thresholds, relative scales, tie rules, and per-layer support counts.

Fixed-density overlap comparisons accompany absolute-threshold masks. A stratified random baseline preserves layer and source-weight-magnitude bins when feasible. Degenerate or tie-dominated gradients are marked; ranking a zero vector does not define a meaningful learned subnetwork. Common-prefix teacher distances are aggregated using the same domain weights, and distance estimates computed from truncated vocabulary outputs are labeled as approximations with their retained mass.

A.2 OPTIONAL ANALYSIS IF THE LOSS COMPARISON SHOWS A STABLE DIFFERENCE

These controls are conditional follow-up, not part of the required PG/top- k experiment. On a fixed student-prefix bank, independently resample next actions under the current student and compare action sample counts, an action-independent detached baseline, and an exact full-vocabulary gradient on a small bank. These known estimator controls can determine whether an observed parameter

effect tracks sampling variance. A retained-head plus sampled-tail estimator can distinguish truncation from the full reverse-KL reference (Oh et al., 2026; Zhang & Ba, 2026; Yu et al., 2026).

The independent action draws do not extend the original response. This keeps prefix selection and weights fixed for the local estimator comparison. Using the original rollout token with its eventual response-length weight need not satisfy the same conditional sampling identity. Forward KL, teacher versus student top- k , or within-set renormalization are objective changes and are labeled separately. These controls are pursued only if they answer a specific difference observed in the primary comparison.

A.3 AN OPTIONAL AMPLITUDE CALIBRATION FOR TEACHER DISTANCE

A simple artificial target can help distinguish update amplitude from parameter selection. Freeze p_0 and q on the same prefixes and set

$$q_\alpha(v | h) = \frac{p_0(v | h)^{1-\alpha} q(v | h)^\alpha}{Z_\alpha(h)}. \quad (11)$$

At the reference student, the full reverse-KL gradient obeys

$$\nabla_\theta D_{\text{KL}}(p_\theta \| q_\alpha)|_{\theta_0} = \alpha \nabla_\theta D_{\text{KL}}(p_\theta \| q)|_{\theta_0}. \quad (12)$$

The log-ratio is an α -scaled version of the original log-ratio plus a prefix-constant normalizer. The constant vanishes under the score identity. Positive scaling therefore preserves fixed-density raw-gradient support while possibly changing absolute-threshold sparsity. This optional calibration does not replace the real-teacher comparisons and does not assert that artificial targets have matched downstream capability.

A.4 REPRODUCIBILITY RECORDS

Record exact student, teacher, tokenizer, data, and evaluator revisions; teacher training lineages; prompt schedules and seeds; decoding, EOS handling, cap, invalid responses, and loss masks; learning-rate schedule, optimizer moments, precision, clipping, and decay; candidate sets, probability normalization, teacher/student retained mass, and advantage handling. Keep per-prompt outputs, source measurements for every plot, and actual time and memory costs.

Distinguish training seeds, diagnostic batch replicates, and evaluation resamples. Pairs sharing a teacher are dependent, and prompt-level uncertainty does not establish population-level teacher relationships. Historical synthetic GPAS calculations remain analytical artifacts and are not empirical evidence for these studies.